Day 125: The Week 18 Build — t-SNE and UMAP From Scratch

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 18, Day 7 | Complete study guide

Watch: StatQuest: t-SNE, Clearly Explained (StatQuest with Josh Starmer) | UMAP explained | The best dimensionality reduction? (AI Coffee Break with Letitia)

Script: `day125_tsne_umap.py` · every number in this guide is copied by program from `day125_run_output.txt`, the real output of that script.

t-SNE and UMAP draw high-dimensional data on a page by keeping each point's neighbours close. Both are written from scratch and land beside the reference libraries: t-SNE's final KL 0.6833 against scikit-learn's 0.6800, UMAP's trustworthiness 0.9880 against umap-learn's 0.9888. The week's six parts, copied in verbatim, do the rest. Clustered with the week's own k-means, the t-SNE map of the digits scores 0.8908, against 0.6495 on ten PCA components. The cautions are measured too: UMAP kept the distances between digit groups worse than t-SNE here (0.2564 for the compact version, 0.5720 for umap-learn, against 0.8232), and the picture depends on perplexity.

0. Where today fits

DEFINITION — start here, no prior knowledge assumed

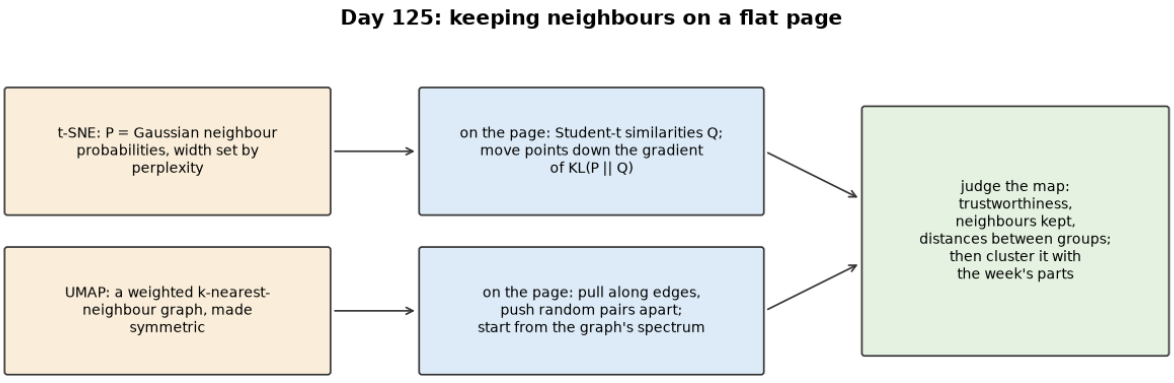
t-SNE (t-distributed stochastic neighbour embedding) turns distances into neighbour probabilities P, places points on a page, and moves them until the page's similarities Q match P, measured by the **KL divergence**. **Perplexity** sets how many neighbours each point listens to.

UMAP (uniform manifold approximation and projection) builds a weighted graph of each point's nearest neighbours, then lays it out by pulling connected points together and pushing random pairs apart. Both keep *local* structure; neither promises that distances between far-apart groups mean anything. Day 125 is the week's **build**: the exercise parts of Days 119 to 124 are copied in unchanged.

REAL-WORLD SCENARIO

A seating plan for a wedding of 200 guests: people who know each other should sit together. The plan keeps friends at the same table, but how far apart two tables are says little about how well their guests know each other.

SYSTEM DIAGRAM



Two ways to keep neighbours on a flat page, judged and clustered with the week's parts.

Data: the 1,797 handwritten digits (64 pixels each), bundled with scikit-learn, labels used only to judge; and 1,000 random points made on purpose. scikit-learn's t-SNE (its exact method) and the umap-learn library are

outside checks.

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
MATHEMATICS	The formula, when the section introduces one.
SPECIFICATION	The exact setup of the experiment.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. Checks: the parts are verbatim

DEFINITION — start here, no prior knowledge assumed

The script parses Days 119 to 124's exercise files and checks that every function appears in itself character for character.

REAL-WORLD SCENARIO

Checking that the parts on the assembly line are exactly the parts that passed inspection.

ACTUAL OUTPUT

```
functions from day119_exercise.py to day124_exercise.py found here, character for character: 26 of
    26
left out as the same code as Day 119's sqdist: sqdist (day120), sqdist (day121)
Actual output from this lesson's run.
```

What it means

All 26 of 26 parts are present: k-means, agglomerative clustering, DBSCAN, the EM steps, PCA and the SVD tools. Days 120 and 121 each carried a copy of Day 119's sqdist; one is enough.

WHY CHOOSE THIS?

The build cannot drift from the parts it claims to use.

WHY NOT THIS? (TRADEOFFS)

A verbatim check proves the copy, not that the parts suit today's maps.

2. t-SNE's pieces

DEFINITION — start here, no prior knowledge assumed

Each point's Gaussian width is found by binary search so that its perplexity is exactly 30; then the KL gradient formula is checked against finite differences.

REAL-WORLD SCENARIO

Tuning each guest's 'circle of friends' to about thirty people, however crowded their corner of the room.

MATHEMATICS

$$p_{j|i} \propto exp(-\beta_i |x_i - x_j|^2),\ \beta_i \text{ set so that } exp(entropy) = perplexity; \quad p_{ij} = (p_{j|i} + p_{i|j})/2n$$

$$q_{ij} \propto (1 + |y_i - y_j|^2)^{-1}; \quad KL = \Sigma\, p_{ij}\, log(p_{ij}/q_{ij}); \quad \partial KL/\partial y_i = 4\, \Sigma_j\, (p_{ij} - q_{ij})(y_i - y_j)(1 + |y_i - y_j|^2)^{-1}$$

IMPLEMENTATION

```
def p_conditional(D2, perplexity, iters=64, tol=1e-6):
    n = len(D2); beta = np.ones(n); lo = np.zeros(n); hi = np.full(n, np.inf); target =
        np.log(perplexity)
    off = ~np.eye(n, dtype=bool)
    E = np.where(off, D2 - np.where(off, D2, np.inf).min(1, keepdims=True), 0.0)
    for _ in range(iters):
        P = np.where(off, np.exp(-E * beta[:, None]), 0.0); s = P.sum(1)
        H = np.log(s) + beta * (P * E).sum(1) / s
        too_flat = H > target
        lo = np.where(too_flat, beta, lo); hi = np.where(too_flat, hi, beta)
        beta = np.where(np.isinf(hi), beta * 2, (lo + hi) / 2)
        if np.abs(H - target).max() < tol:
            break
    return P / s[:, None], np.exp(H)
def tsne_grad(Y, P):
    num = 1.0 / (1.0 + np.maximum(sqdist(Y, Y), 0)); np.fill_diagonal(num, 0)
    Q = np.maximum(num / num.sum(), 1e-12); PQ = (P - Q) * num
    grad = 4 * (PQ.sum(1)[:, None] * Y - PQ @ Y)
    return float((P * np.log(P / Q)).sum()), grad
```

ACTUAL OUTPUT

```
digits: every point's perplexity after the binary search: 30.0000 to 30.0000 (target 30); P sums
    to 1.000000
40 digits, a random layout: KL 1.6501; the gradient formula against finite differences, largest
    gap 2.7e-10 (largest component 0.033)
Actual output from this lesson's run.
```

What it means

Every one of the 1,797 points ends at perplexity 30.0000, and P sums to 1. The gradient formula agrees with finite differences to 2.7e-10: the optimiser follows the true slope of the KL divergence.

WHY CHOOSE THIS?

Each piece is checked before the whole is trusted.

WHY NOT THIS? (TRADEOFFS)

A correct gradient does not guarantee a good optimum; t-SNE's KL has many.

3. t-SNE on the digits, against scikit-learn

DEFINITION — start here, no prior knowledge assumed

1,000 steps: early exaggeration (P x 12) for 250, momentum, per-coordinate gains, and Day 123's PCA as the start, as scikit-learn does.

REAL-WORLD SCENARIO

Two cooks following the same recipe in different kitchens.

IMPLEMENTATION

```
def tsne(X, perplexity=PERPLEXITY, iters=1000, seed=0, exaggeration=12.0, exag_iters=250):
    n = len(X); P, _ = joint_p(X, perplexity); lr = max(n / exaggeration / 4, 50)
    mu, vals, V = pca_fit(X); Y = transform(X, mu, V, 2); Y = Y / Y[:, 0].std() * 1e-4
    upd = np.zeros_like(Y); gains = np.ones_like(Y); kl = []
    for t in range(iters):
        ex = exaggeration if t < exag_iters else 1.0
        _, g = tsne_grad(Y, P * ex)
        gains = np.where(upd * g < 0, gains + 0.2, gains * 0.8).clip(0.01)
        upd = (0.5 if t < exag_iters else 0.8) * upd - lr * gains * g
        Y = Y + upd
        if (t + 1) % 50 == 0:
            kl.append(tsne_grad(Y, P)[0])
    return Y, kl
```

ACTUAL OUTPUT

```
                 final KL  trustworthiness  seconds
  here             0.6833           0.9926    126.9
  scikit-learn     0.6800           0.9923     49.9
KL every 200 steps here: [2.6847, 0.8737, 0.735, 0.6986, 0.6833]; early exaggeration ends at step 250
trustworthiness written here against scikit-learn's function, on this map: 0.992596 against 0.992596
Actual output from this lesson's run.
```

What it means

The two reach almost the same KL (0.6833 and 0.6800), and their maps are equally trustworthy (0.9926 and 0.9923). The trustworthiness written today matches scikit-learn's own function to six decimals (0.992596). This version is slower (126.9 s against 49.9 s in this run): it computes all n^2 pairs with whole-matrix operations every step.

WHY CHOOSE THIS?

A faithful t-SNE in about 35 lines.

WHY NOT THIS? (TRADEOFFS)

Every step costs n^2; libraries approximate (Barnes-Hut) for large n.

4. UMAP on the digits, against umap-learn

DEFINITION — start here, no prior knowledge assumed

A compact UMAP: the k-nearest-neighbour graph with UMAP's weights, the curve $1/(1 + a\,d^{2b})$ fitted by a grid search, a spectral start, and 500 epochs of pulling and pushing, applied to all edges at once rather than one edge at a time.

REAL-WORLD SCENARIO

Building a map of friendships from who stands next to whom, then shaking it until friends sit together and strangers drift apart.

IMPLEMENTATION

```
def fuzzy_graph(X, k=NEIGHBOURS):
    n = len(X); D = np.sqrt(np.maximum(sqdist(X, X), 0)); np.fill_diagonal(D, np.inf)
    idx = np.argsort(D, 1)[:, :k]; d = np.take_along_axis(D, idx, 1)
    rho = d[:, 0]; lo = np.zeros(n); hi = np.full(n, np.inf); sig = np.ones(n); target =
        np.log2(k)
    for _ in range(64):
        s = np.exp(-np.maximum(d - rho[:, None], 0) / sig[:, None]).sum(1)
        big = s > target
        hi = np.where(big, sig, hi); lo = np.where(big, lo, sig); sig = np.where(np.isinf(hi), sig
            * 2, (lo + hi) / 2)
    W = np.zeros((n, n)); W[np.arange(n)[:, None], idx] = np.exp(-np.maximum(d - rho[:, None], 0)
        / sig[:, None])
    return W + W.T - W * W.T
```

ACTUAL OUTPUT

```
the curve 1 / (1 + a d^(2b)): a 1.5811, b 0.8950 here (a grid search); umap-learn's fit: a 1.5769, b 0.8951
                trustworthiness  10 nearest kept  seconds
  here                   0.9880           0.4775      3.1
  umap-learn             0.9888           0.4913      9.3
(umap-learn's time includes compiling its code with numba on first use)
Actual output from this lesson's run.
```

What it means

The curve matches umap-learn's (1.5811 and 0.8950 against 1.5769 and 0.8951); the small gap in a is the grid's step. The maps are close in trustworthiness (0.9880 and 0.9888) and in neighbours kept (0.4775 and 0.4913). This version took 3.1 s; umap-learn's 9.3 s includes compiling its code on first use.

WHY CHOOSE THIS?

UMAP's ideas in a page of NumPy, and fast on a small dataset.

WHY NOT THIS? (TRADEOFFS)

Dense n x n matrices; real UMAP uses approximate neighbour search and sparse graphs.

5. Three maps compared

DEFINITION — start here, no prior knowledge assumed

PCA (2 components), t-SNE and UMAP, judged on each point's neighbourhood and on the distances between the ten digit groups' centres.

REAL-WORLD SCENARIO

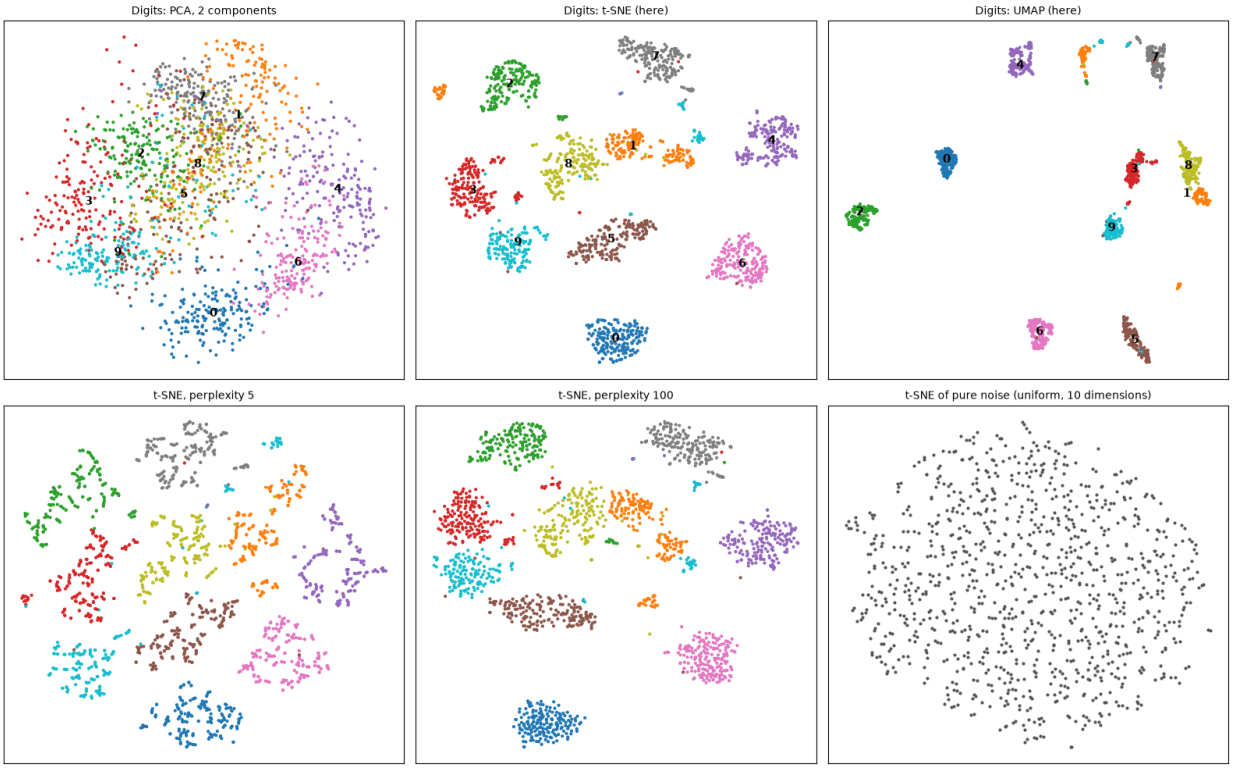
Three seating plans for the same wedding: one by height, two by friendship.

IMPLEMENTATION

```
def trust(X, Y, k=10):
    n = len(X); _, DX = knn_sets(X, k); ranks = np.argsort(np.argsort(DX, 1), 1) + 1
    NY, _ = knn_sets(Y, k); r = np.take_along_axis(ranks, NY, 1)
    return float(1 - 2 / (n * k * (2 * n - 3 * k - 1)) * np.maximum(r - k, 0).sum())
def kept(X, Y, k=10):
    NX, _ = knn_sets(X, k); NY, _ = knn_sets(Y, k)
    return float(np.mean([len(set(a) & set(b)) / k for a, b in zip(NX, NY)]))
def centre_corr(X, Y, y):
    cx = np.array([X[y == c].mean(0) for c in range(10)]); cy = np.array([Y[y == c].mean(0) for c
        in range(10)])
    iu = np.triu_indices(10, 1); return float(np.corrcoef(np.sqrt(sqdist(cx, cx)[iu]),
        np.sqrt(np.maximum(sqdist(cy, cy), 0)[iu]))[0, 1])
```

ACTUAL OUTPUT

```
  map                    trustworthiness  10 nearest kept  digit-centre distances (corr)
  PCA, 2 components               0.8300           0.1179                        0.8195
  t-SNE (here)                    0.9926           0.5825                        0.8232
  UMAP (here)                     0.9880           0.4775                        0.2564
  UMAP (umap-learn)               0.9888           0.4913                        0.5720
trustworthiness and 'kept' judge each point's neighbourhood; the last column, distances between the 10 digit
    groups' centres
Actual output from this lesson's run.
```



Top: the digits by PCA, t-SNE and UMAP, coloured by the true digit (numbers at each group's middle). Bottom: t-SNE at perplexity 5 and 100, and t-SNE of pure noise.

What it means

PCA keeps the large-scale layout (0.8195 correlation of group distances) but mixes neighbourhoods: only 0.1179 of each point's ten nearest stay nearest. t-SNE keeps both here (0.5825 and 0.8232), probably helped by its PCA start (not tested). UMAP keeps neighbourhoods nearly as well, but its groups' distances follow the original much less (0.5720 for umap-learn, 0.2564 for the compact version, which differs from it in start and update details). The common claim that UMAP keeps global structure better than t-SNE did not hold on this dataset.

WHY CHOOSE THIS?

Measures instead of impressions.

WHY NOT THIS? (TRADEOFFS)

One dataset; the global comparison depends on how each method is started.

6. Clustering the maps with the week's parts

DEFINITION — start here, no prior knowledge assumed

The week's four clusterers, k = 10, on ten PCA components and on the two maps. DBSCAN's eps is fixed in advance: the 90th percentile of the distance to the 10th nearest point (Day 121's k-distance).

REAL-WORLD SCENARIO

Drawing table boundaries on a seating plan, rather than on the guest list.

ACTUAL OUTPUT

```
  representation                  k-means             Ward  Gaussian mixture            DBSCAN   DBSCAN
      noise
  PCA, 10 components               0.6495           0.6977            0.7325            0.0001         0.010
  t-SNE (here)                     0.8908           0.8861            0.8859            0.8811         0.016
  UMAP (here)                      0.8272           0.8389            0.8272            0.8197         0.026
k-means: Day 119; Ward: Day 120; Gaussian mixture: Day 122's E and M steps; DBSCAN: Day 121, eps from Day
    121's k-distance; PCA: Day 123
Actual output from this lesson's run.
```

What it means

Every clusterer does far better on the maps: k-means 0.8908 on t-SNE against 0.6495 on PCA, and DBSCAN goes from useless (0.0001) to 0.8811. On the maps the digits are already compact islands, so any reasonable clusterer finds them. The catch: the islands were drawn by t-SNE or UMAP, and their shapes and gaps are partly the method's doing, so clustering a map is a way to read the picture, not an independent discovery.

WHY CHOOSE THIS?

A practical recipe: embed, then cluster.

WHY NOT THIS? (TRADEOFFS)

Results inherit every choice made in the embedding, including perplexity (section 7).

7. Warnings: perplexity, and pure noise

DEFINITION — start here, no prior knowledge assumed

t-SNE at perplexity 5, 30 and 100; then 1,000 points spread evenly through a 10-dimensional cube, where there are no groups at all.

REAL-WORLD SCENARIO

Asking a skilled sketch artist to draw a crowd that has no groups in it, and checking whether the drawing shows any.

ACTUAL OUTPUT

```
   perplexity  trustworthiness  k-means ARI  digit-centre corr
            5           0.9911       0.8204             0.7947
           30           0.9926       0.8908             0.8232
          100           0.9915       0.8199             0.8153
pure noise (1,000 uniform points in 10 dimensions), k-means with k = 5; silhouette near 0 means no
    real groups:
  the points themselves : silhouette 0.0777
  PCA, 2 components     : silhouette 0.3232
  t-SNE map             : silhouette 0.3531
any flat 2-D cloud scores higher than a 10-D one; t-SNE is judged against PCA's 2-D picture of the
    same points
Actual output from this lesson's run.
```

What it means

Perplexity changes the picture more than the scores. At 5 the digits break into fragments (bottom left of the figure) and k-means falls to 0.8204; at 100 the groups are rounder but k-means again scores lower (0.8199); 30 was best here (0.8908). Trustworthiness barely moves (0.9911 to 0.9926): every setting keeps neighbourhoods, while the arrangement of groups changes.

On pure noise, the common warning that t-SNE invents clusters did not hold here. The map is an even disc (bottom right), and its k-means silhouette, 0.3531, is close to that of PCA's flat picture of the same points (0.3232). Both are far above the points' own 0.0777 largely because any 2-D cloud scores higher than a 10-D one. Smaller perplexities and other kinds of noise were not tested.

WHY CHOOSE THIS?

Shows what the pictures can and cannot be trusted for.

WHY NOT THIS? (TRADEOFFS)

One noise test at one perplexity; the warning may still apply in other settings.

Interview preparation — technical questions

Q: t-SNE, UMAP or PCA for a picture of high-dimensional data?

A: t-SNE or UMAP for neighbourhoods (trustworthiness 0.99 against PCA's 0.83); PCA when large-scale distances matter.

Q: What does perplexity control?

A: How many neighbours each point listens to; 5 fragmented the digits, 30 scored best here.

Q: Can you cluster a t-SNE map?

A: It works well here (k-means 0.8908), but the clusters partly reflect the method's choices.

Q: How do you check a t-SNE implementation?

A: Calibrated perplexities, a finite-difference gradient check (2.7e-10), and KL against a reference (0.6833 against 0.6800).

Q: Why is UMAP faster?

A: It only looks at k nearest neighbours and random pairs, not all n^2 pairs.

Interview preparation — theoretical questions

Q: Why the Student-t on the page?

A: Its heavy tails give room: moderately distant points can sit far apart without large cost.

Q: What is early exaggeration for?

A: Multiplying P for the first 250 steps lets groups form before the fine arrangement.

Q: Do distances between t-SNE clusters mean anything?

A: Not reliably; here they happened to follow the original (0.8232), probably helped by the PCA start.

Q: What does trustworthiness measure?

A: Whether a point's neighbours on the map were really near it: map neighbours that were far cost by their original rank.

Key takeaway and what's next

> **KEY TAKEAWAY**
> t-SNE and UMAP keep what PCA cannot: who is next to whom. Written from scratch, both matched the reference libraries closely, and their maps made the week's simplest clusterers look excellent on the digits. The cautions are part of the result: the picture depends on perplexity and on how the method is started, and distances between groups on the page are not a measurement.

- t-SNE: KL 0.6833 (scikit-learn 0.6800); trustworthiness 0.9926.
- UMAP: trustworthiness 0.9880 (umap-learn 0.9888).
- k-means on the t-SNE map 0.8908, on ten PCA components 0.6495.
- Group distances kept: t-SNE 0.8232, umap-learn 0.5720, PCA 0.8195.

Suggested watch order, if starting from zero

- 1. StatQuest: t-SNE, Clearly Explained (StatQuest with Josh Starmer) — t-SNE's idea, step by step.
- 2. UMAP explained | The best dimensionality reduction? (AI Coffee Break with Letitia) — UMAP's graph and layout, and how it differs from t-SNE.

Day 126 preview

- **Week 19 begins: autoencoders from scratch**, with a real reconstruction demonstration.